EXECUTIVE SUMMARY

Public secondary school students across five pilot states (Zamfara, Ebonyi, Kebbi, Cross River, and Yobe) face severe educational inequities due to extreme learning resource shortages, localized information silos, and prohibitive data costs. This is verified by a low 42% rural WAEC pass rate (compared to 71% in urban areas), which blocks higher education access and allows urban applicants to capture 68% of available scholarships, reinforcing a multi-generational cycle of rural poverty.

Implement Option C: The EduBridge Full Platform. This integrated, offline-first digital enterprise solution combines an offline content library, scholarship matching with automated SMS alerts, mentor connect, a teacher portal, and a parent SMS channel deployed via 500 low-cost Android tablets to establish shared media labs across 50 public secondary pilot schools.

The initiative requires a total initial capital investment (CapEx) of ₦108,000,000 for one-time engineering and deployment, alongside a fixed annual operating expenditure (OpEx) run-rate of ₦17,600,000. Over a three-year window, the platform will systematically unlock ₦1.5 billion in direct higher education scholarship funding for rural students and deliver ₦280,800,000 in monetized teacher time-savings. It achieves standalone financial self-sustainability by Year 3 (generating ₦75,000,000 in subscription and partnership revenue against the ₦17.6M running cost) and injects ₦90 billion in graduate lifetime earnings premium into the rural economy.

Full pilot delivery is bound to a strict 10-month critical path, moving sequentially from foundational setups and software configurations to user training and live field deployment

Solution Options Considered

Option A Do nothing

Continue with the current situation: no learning support platform, no scholarship directory, no mentor matching. Rural students remain disadvantaged. Calculate the annual social and economic cost of inaction: 68% of scholarship slots continue to go to urban students (₦X billion in foregone rural scholarships annually). Knowledge gap compounding effect on WAEC pass rates (rural WAEC pass rate: 42% vs urban: 71% — Delta Research Institute, 2022). Graduate earnings impact of unearned scholarships and unmatched mentors

Option B : Printed scholarship directory with SMS alerts

A lower-cost option addressing scholarship awareness only. Printing and distributing a directory to the 50 pilot schools costs roughly ₦5,000,000 per run; because scholarship terms change through the year, at least two runs a year are needed, putting recurring cost at around ₦10,000,000 annually before distribution to schools 100–300km from state capitals. It does nothing for the learning-support, mentor-access, teacher-development or parent-engagement gaps identified during requirements gathering, and it is out of date the moment a single scholarship changes its deadline; a database can be corrected within 24 hours; a print run cannot.

Option C: Full EduBridge platform (recommended)

An offline-first platform combining an offline content library, scholarship matching with automated SMS alerts, mentor connect, a teacher portal, and a parent SMS channel. It is the only option that addresses all

five root problems identified in the stakeholder and root-cause analysis, and the only one built to work without a personal data connection, the binding constraint for the primary user group. Because SMS and offline-cached content carry near-zero marginal cost per additional student once built, unit cost falls as the pilot scales, which Option B's linear per-copy cost cannot do.

Recommendation: Option C.

Cost-Benefit Analysis

One-time Costs

Cost item	**Amount (#)**
Platform development (5 modules)	48,000,000
WAEC-aligned content creation	22,000,000
Scholarship database set-up	3,000,000
Mentor recruitment and onboarding	4,000,000
Device provision (500 tablets)	25,000,000
Teachers training (750 Teachers)	6,000,000
Total One-time costs	108,000,000

Annual Recurring Cost

Cost Item	**Amount (#)**
Server hosting, maintenance & SMS	8,000,000
Operation team (6 staff)	9,600,000
Total Recurring Costs (per year)	17,600,000

Year 1 total cost: ₦125,600,000

1. **Benefit summary**

Benefit stream	**Year 1 Amount (#)**	**Year 2 Amount (#)**	**Year 3 Amount (#)**
Scholarship funding unlocked	100,000,000	400,000,000	1,000,000,000
Government performance grant	0	30,000,000	0
School subscription	0	30,000,000	60,000,000
Scholarship body	0	10,000,000	15,000,000

co-funding			
Teachers time savings (monetized)	93,600,000	93,600,000	93,600,000
Total Benefits	**193,600,000**	**563,600,000**	**1,168,600,000**

Financial Return on Investment (ROI) Calculations

Year 1 Financial ROI:

Net Benefit = Year1 Benefit - Year1 Total Cost

Net Benefit = ₦193,600,000 − ₦125,600,000 = ₦68,000,000.

ROI = (₦68,000,000 ÷ ₦125,600,000) × 100 = 54.1%.

Three-year Cumulative ROI

Three-year total costs = One Time Cost + (3 years Recurring Cost)

Three-year total costs = ₦108,000,000 + (₦17,600,000 × 3) = ₦160,800,000.

Three-year total benefits = ₦193,600,000 + ₦563,600,000 + ₦1,168,600,000 = ₦1,925,800,000.

Net benefit = ₦1,925,800,000 − ₦160,800,000 = ₦1,765,000,000.

ROI = (₦1,765,000,000 ÷ ₦160,800,000) × 100 = 1,097.6%.

Social Return on Investment (SROI)

Based on World Bank macroeconomic human development datasets, a rural student who clears academic exam hurdles, accesses a ₦200,000 university scholarship, and finishes higher education secures a projected **lifetime earnings premium of ₦12,000,000 to ₦18,000,000** relative to terminal secondary school dropouts

Year	**Additional scholarship students**	**Lifetime earnings premium (#12m)**
Year 1	500	6,000,000,000
Year 2	2,000	24,000,000,000
Year 3	5,000	90,000,000,000

Three-year social value: ₦90 billion in additional lifetime earnings potential. This is a conservative value.

Social ROI Ratio = ₦90,000,000,000 ÷ ₦160,800,000 ≈ 560:1.

That means every ₦1 invested in EduBridge is projected to generate approximately ₦560 in long-term social and economic value.

1. **Cost Per Student Reached**

Pilot Cost Efficiency

Year 1 cost = #125,600,000
Pilot Cohort = 25,000 students
Cost per student (year 1) = #125,600,000 ÷ 25,000 students = 5,024
Cost per student = **5,024**

2. **Conclusion (Cost Benefit Analysis Verdict)**

Financial Verdict

- Year 1 ROI: 54.1% (positive within the first year)
- Three-year ROI: 1,097.6%
- Cost per student: #5,024 for the 25,000 students pilot Cohort

Social Impact Verdict

- Conservative three-year social value: 90,000,000,000
- Social ROI ratio: ~560:1
- Directly addresses the documented inequities in textbook access, scholarship awareness, and mentorship for rural students.

RISK REGISTER

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy	Contingency Plan
R-001	Device theft or damage: Shared tablets in under-resourced public schools are high-theft items. Without devices, the offline content library cannot be accessed by students without personal phones	High	High	Deploy asset tracking barcodes; install steel lockable storage boxes in the principal's office; enforce signed school asset registries	Access third-party insurance pools; open a lightweight web-browser fallback application for students with parental Android phones.
R-002	Teachers who do not actively recommend EduBridge to students and who do not enter quiz scores mean the progress tracking and parent SMS features are empty. The platform becomes invisible.	High	Medium	Conduct a mandatory 2-day teacher onboarding training program; integrate platform usage directly with the Ministry's continuous assessment metrics.	Run bi-weekly data-sync audits; deploy field engagement officers to run onsite coaching; activate automated principal dashboards.

R-003	Content quality disputes: If the offline learning content has errors, or is not aligned to the current WAEC syllabus, students who rely on it may perform worse than if they had studied from their teachers' notes.	Low	High	Contract certified WAEC subject-matter specialists to author all course content; implement a rigorous multi-stage technical proofing board before coding.	Execute annual syllabus validation sweeps; push rapid over-the-air digital patch files or deploy flash-drive updates during monthly field syncs.
R-004	Mentors Accountability: If volunteer mentors accept match requests and then do not appear for sessions, students who have never had a mentor before experiencing rejection. This is psychologically damaging and destroys trust in the platform.	Medium	High	Require mentors to complete 3 video orientation modules; enforce signed commitment agreements; deploy automated 24-hour and 2-hour dashboard reminders.	Maintain a dedicated, rotating on-call standby pool of backup mentors to step into unfulfilled live match sessions in real time
R-005	Scholarship database becomes outdated: If a scholarship deadline or requirement changes and EduBridge's database is not updated, a student who prepared and applied according to the wrong deadline loses their chance.	Medium	High	Assign 1 full-time internal analyst to manual database validation; build automated custom web-scrapers to monitor partner sponsor portals.	Disseminate immediate emergency bulk SMS warning alerts to affected students and parent contacts the moment an external deadline shifts.
R-006	Student safeguarding risk: A mentor uses the platform to make inappropriate contact with a student under 16. NDPR and child protection law implications are severe.	Low	Critical	Route all messages through automated anonymization proxies; block raw phone numbers/social	Program algorithm-driven automated text chat filters for flags; execute lifetime profile termination; hand conversation logs directly to law

				handles; mandate API-linked National Identification Number (NIN) checks	enforcement.
R-007	State Ministry mandate creates backlash: If a State Ministry mandates all schools to use EduBridge, voluntary adoption becomes forced compliance and teacher resistance intensifies.	Medium	Medium	Conduct collaborative co-design workshops with local teacher unions early; pitch the software utility tools as work-savers, not administrative burdens.	Readjust rollout messaging to focus heavily on teacher administrative efficiency gains (3 hours per week saved from chalkboard transcription
R-008	If the pan-African foundation delays the ₦200M grant disbursement, the project cannot start on schedule and the pilot school intake date is missed.	Low	High	Finalize and submit all required compliance documentation early; lock down explicit tranche disbursement milestones in the grant contract.	Secure an interim bridge financing credit line with partner social impact banks using the signed foundation grant commitment as collateral.

Implementation Roadmap

Phases & Timeline	Milestones	Key Dependencies:	High-level Delivery Plan
Phase 1 — Foundation *(Months 1–2)*	Foundation grant disbursement received; 50 scholarships verified and loaded; 200 mentors recruited and briefed; all 50 school principals sign participation agreements.	Timely legal clearance of funding tranches by the foundation board to trigger initial project spending.	Finalize requirements sign-off with the Executive Director; complete foundation grant paperwork; execute curriculum specialist content commissions for 4 subjects; run scholarship database research for 50 scholarships; launch the mentor recruitment campaign; hold pilot school principal briefings across 5 states.
Phase 2 — Build *(Months 3–6)*	All modules pass technical QA on the lowest-spec target hardware (Android Go, 1GB RAM); complete comprehensive prototype demo delivered to the Executive Director and foundation board representative	Keeping local software data packets lean to run seamlessly without continuous network access.	Code the offline content library engine (WAEC 4 subjects); develop the scholarship matching database filters and mentor matching algorithms; program the teacher portal, parent SMS configuration system, student progress dashboards, and run low-end Android device optimization testing.
Phase 3 — Content & Training *(Month 7)*	750 public school teachers certified; all 500 hardware tablets provisioned and active; all 50 schools complete student onboarding; 5,000 core students registered by the end	Regional logistics clearance and hardware security configuration arrays at the pilot installations.	Coordinate formal content quality reviews with certified WAEC subject-matter specialists; execute all 750 teacher training workshops via a 2-day program per state across 5 separate

	of Month 7.		training events; manage device provisioning to all 50 schools (10 tablets per school lab); run school-level student onboarding orientations.
Phase 4 — Pilot Launch *(Months 8–9)*	25,000 active students registered; maintain a 60%+ Monthly Active User (MAU) metric; first 100 scholarship applications submitted through direct EduBridge guidance; first 500 mentor sessions completed.	Continuous network availability of the core bulk cellular SMS gateway.	Launch full operational rollout across all 50 pilot locations; monitor daily user engagement metrics; initiate automated outbound scholarship deadline alerts; coordinate first-wave mentor matches; conduct weekly teacher support calls.
Phase 5 — Foundation Review *(Month 10)*	Foundation 6-month impact report formally approved; board decision secured for national expansion to 500 schools; commercial school subscription billing model operational.	Favorable data correlation across student exam pass metrics.	Compile and submit the formal 6-month impact report to the foundation; perform data analysis on student learning progress, scholarship submission trends, and mentor engagement analytics; host the foundation evaluation review board meeting; launch the school subscription billing infrastructure for non-grant expansion schools.

SUCCESS METRICS (KPIs)

Beneficiary KPIs

- Customer/User Satisfaction Score (CSAT)

- Net Promoter Score (NPS)
- Student Platform Retention Rate

Educational Operational KPIs

- Average Lesson Prep-Time Savings
- Scholarship Application Completion Rate
- WAEC Examination Accuracy Rate

Social Enterprise Financial KPIs

- Independent Monthly Recurring Revenue (MRR)
- Ecosystem Lifetime Social Capital Value
- Annual Institutional Revenue Growth Rate